JAMES GRAY

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EDUCATION

University of Warwick – BEng Computer Systems Engineering (Year in Industry) Sept. 2022 – June 2026

- **Grade: First (83.3%)**
- **Key Modules:** AI, Data Structures, Operating Systems & Networks, Data Analytics, Computer Architecture, FPGAs, Software Engineering, Maths

Reading School Sept. 2020 – July 2022

- **Grades: A*A*A*** (Maths, Physics, Computer Science)

SKILLS & AWARDS

- **Languages:** C, C++, C#, Python, Java, Haskell, Verilog, MATLAB
- **Web & Data:** React, TypeScript, JavaScript, HTML, CSS, SQL
- **Awards:** Exceptional Achievement in Second Year, Computer Science Department Award

WORK EXPERIENCE

Qube Research & Technologies – Quantitative Technology Intern June 2024 – Present

- Designed and built a **C++ process-management service** that lets support teams control and monitor system processes via a UI and code, eliminating manual SSH interaction.
- Developed a tool that **auto-generates environment-specific configurations** and wires them into the process-management service, establishing a single source of truth.
- Reworked the **CI/CD pipeline** to support **C# builds and tests on Linux**, raising code coverage and unblocking Linux developers.
- Built an **LLM support assistant** (React.js) on a **ReAct loop** with custom tools for multi-step reasoning, context access, and output verification, improving answer accuracy.

MyTutor – Tutor Jan. 2023 – Present

- Communicated complex concepts simply to help **A-Level and GCSE students** master subject material.

KEY PROJECTS

Conjugate Gradient Optimisation on a 3D Mesh (C, AVX-256, OpenMP, Cache Optimisation)

- Optimised a 3D-mesh conjugate gradient solver with **AVX-256**, **OpenMP**, and locality refactoring for an **8.88x speedup**.

Multithreaded Packet Sniffer (C, Networking, Multithreading)

- Detects domain blacklist violations, SYN floods, and ARP cache poisoning.
- Implemented a thread-safe packet queue with **pthreads** feeding a thread pool, processing **1,000,000s of packets without loss**.
- Validated for memory leaks and race conditions with **Valgrind, Helgrind, and GDB**.

FPGA Pacman (Verilog, Vivado)

- A recreation of Pacman on a Nexys 4 FPGA; achieved the **top score in the year**.
- Implemented **custom VGA drivers** for graphics output, including frame-buffer control and pixel-timing generation.
- Debugged signal timings across digital modules using **Verilog test benches in Vivado**.

Resistor Image Scanner (Python, OpenCV, Flask)

- A web app that reads resistor values from photos using image processing; scored **100%**.
- Built **OpenCV** pipelines for resistor localisation, normalisation (denoising, deblurring, glare removal), and colour extraction.
- Applied **K-means clustering** to segment colour regions and locate resistor bands.

Magnetic Electron Trap Simulation (Python, SciPy, NumPy)

- Modelled and visualised an electron in a magnetic field by solving its differential equations; scored **100%**.
- Parallelised the simulation with **multiprocessing** to bypass the **GIL**, achieving a **20x speedup**.

Other Projects

- Movie Information Viewer (Java, from-scratch data structures), Gig Management System (Java, PostgreSQL), Discord Drive cloud storage (Python, Quart), Rhythm-game Discord bot, Maze & Circuit Solvers, Line-Following Buggy (C), Self-Balancing Robot (Simulink).